

Analysis of Grid Segments and Elements - TGI Use Case Notes

[UC23] - Awesense Platform Use Case



Overview

This manual is intended to:

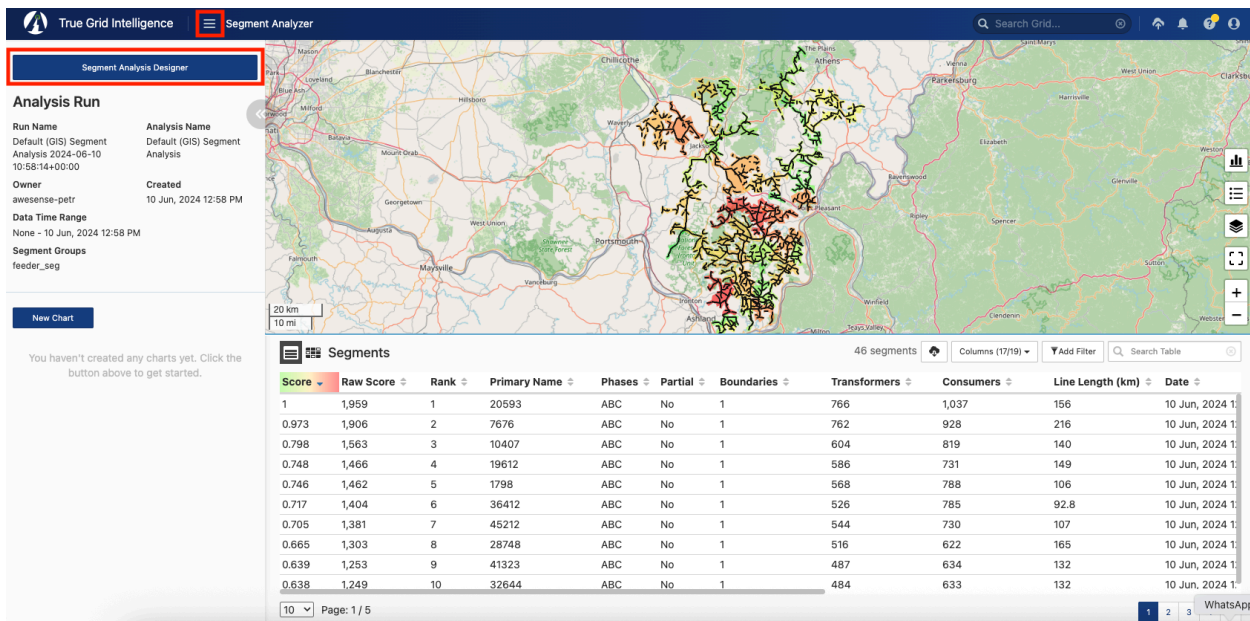
- Provide instructions to create the selected Use case in True Grid Intelligence (TGI).

For more details about the Analysis of Grid Segments and Elements and how it can be analyzed using Awesense Platform, please refer to the [UC23 - Analysis of Grid Segments and Elements.pdf](#) document.

TGI Use Case Notes

This document shows examples of using this Use Case along with a description of how to create such a Use Case on your data and screenshots directly from the TGI visualization interface.

Segment Analyzer can be found directly in TGI in the main menu (select from the main menu - top hamburger icon).



Analysis of Grid's Segments:

To create an Analysis, click on "Segment Analysis Designer".

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Analysis of Grid's Segments or Elements in the Awesense Platform is in general composed of four steps:

1. Create and upload analysis data sources,
2. Creation of analysis factors,
3. Creation of analysis,
4. Running the analysis.

Segment Analysis Designer

RunsAnalysesFactorsData Sources

Analysis Runs ?

10 runsColumnsAdd FilterSearch Table

Run Name	Analysis Name	Data Start Time	Data End Time	Owner	Created
Default (GIS) Segment Analysis 2024-06-10 10:58:14+00:00	Default (GIS) Segment Analysis		10 Jun, 2024 12:58 PM	awesense-petr	10 Jun, 2024
EV Feeder Penetration Analysis 2021-08-23 07:14:59+00:00	EV Feeder Penetration Analysis	18 Jun, 2020 3:00 AM	23 Jul, 2020 8:00 PM	awesense-petr	23 Aug, 2020
EV Feeder Penetration Analysis 2021-05-20 11:31:02+00:00	EV Feeder Penetration Analysis	24 Jul, 2018 12:00 AM	3 Oct, 2020 12:00 AM	awesense-petr	20 May, 2021
EV Feeder Penetration Analysis 3 2020-11-20 22:08:51+00:00	EV Feeder Penetration Analysis 3		20 Nov, 2020 11:08 PM	awesense-andrew	20 Nov, 2020
Grid Wide Feeder Congestion 2020-09-16 23:41:33+00:00	Grid Wide Feeder Congestion	24 Jul, 2018 12:00 AM	24 Aug, 2020 10:00 PM	awesense-andrew	17 Sep, 2020
Test Segment Analysis 2020-09-03 16:08:43+00:00	Test Segment Analysis	27 Jun, 2017 4:00 AM	1 Jan, 2018 12:59 AM	awesense-jakub	3 Sep, 2020
Energy balance - per feeder	Theft Segment Analysis	27 Nov, 2018 1:00 AM	5 Dec, 2018 12:00 AM	awesense-tamir	6 Dec, 2018
Energy balance - opportunistic	Theft Segment Analysis	27 Nov, 2018 1:00 AM	5 Dec, 2018 12:00 AM	awesense-tamir	6 Dec, 2018
All Feeders Ranked by Total Outages (2017) 2018-11-08 18:54:54+00:00	All Feeders Ranked by Total Outages	1 Jan, 2017 1:38 AM	1 Jan, 2018 3:03 PM	michael	8 Nov, 2018
Default (GIS) Segment Analysis 2018-11-08 18:50:40+00:00	Default (GIS) Segment Analysis	30 Apr, 2016 12:27 PM	8 Nov, 2018 7:50 PM	michael	8 Nov, 2018

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Delete RunView Run Results

In the new window, select "Factors".

For the analysis, it is necessary to create a Factor. It is possible to use one of the already created factors (on the left side) or you can create a new factor using the "New Factor" button. For the creation of new factor, the factor's data need to be uploaded to TGI.

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Segment Analysis Designer Runs Analyses **Factors** Data Sources ×

Factors ?

Filter ⊞

Absolute Value Metered Minus Billed

Active Ah Delivered

Active Ah Discrepancy

Active Ah Discrepancy %

Ah Delivered

Ah Discrepancy

Ah Discrepancy %

Consumers

kWh Delivered

New Factor Save Cancel

Identifier

Mostly for bookkeeping. Use underscores instead of spaces.

Display name

Description

Segment Factor Formula

Normalization

If you are creating a new factor, you select a name etc. The calculation is set in the "Segment Factor Formula" section:

- transformation is the way of calculating the data - for example, median, sum etc.
- segment data sources are the uploaded data to be calculated. Some of them are already prepared, such as the length of the line, and others can be uploaded using a CSV file (there is a "Data Sources" button in the top menu of this window).

Then it is necessary to create an Analysis - in the "Analyses" section. It is possible to use one of the already created Analyses (on the left side) or you can create a new analysis using the "New Analysis" button.

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Segment Analysis Designer Runs **Analyses** Factors Data Sources ×

Analyses ?

Filter ⊞

All Trends Analysis

Billing Errors

Default (GIS) Segment Analysis

Demo Segment Analysis 1

Demo Segment Analysis 2

Grid Monitoring Segment Analysis

Grid Optimizing Analysis - Equal Weights

Grid Optimizing Analysis - Outage Heavy

Grid Optimizing Analysis - Theft Heavy

New Analysis Save Cancel

Identifier

Identifier

Mostly for bookkeeping. Use underscores instead of spaces.

Display name

Display name

Description

Description

Segment Score Formula

No Segment Factors Selected

Select Segment Factor ▼

Weight

Add Factor

☐ Allow factors with different normalization schemes ?

If you are creating a new analysis, you select a name etc. It is then necessary to select a factor using the "Select Segment Factor" button and determine its weight. Weight will be applied if we have multiple factors. Finally, click on "Add Factor" and "Save".

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Segment Analysis Designer

RunsAnalysesFactorsData Sources

Analyses ?

Filter

All Trends Analysis

Billing Errors

Default (GIS) Segment Analysis

Demo Segment Analysis 1

Demo Segment Analysis 2

Grid Monitoring Segment Analysis

Grid Optimizing Analysis - Equal Weights

Grid Optimizing Analysis - Outage Heavy

Grid Optimizing Analysis - Theft Heavy

Legacy Segment Analysis

Line_length_per_transformer

Meters kWh for each transformer

NTS Segment Analysis

Number of customers for transformer

Analysis Name: Meters kWh for each transformer

New Analysis

Create Copy

Description
--

Segment Score Formula
(1 * Meters kWh)

Analysis Run Settings

How the calculator should handle data overlaps

☒ Intersection (takes data from the time period where all segment data sources overlap)

☐ Union (takes data from the entire period for which there is any data available for any of the segment data sources)

1 segment groups 0 segments selected

SEGMENT GROUP NAME	CHILD SEGMENTS
All energy balance segments generated in the batch	0

Choose Segment Groups

Run Analysis

Finally, it is necessary to run this analysis. It is necessary to select one of the analyses from the left side and the run is done in the "Analysis Run Settings" section at the bottom of the window, where you select How the calculator should handle data overlaps and select the grid for analysis using the "Choose Segment Groups" button. Here the segment created using UC22 - Topological Grid Segmentation is selected. Finally, the "Run Analysis" button is clicked.

Analysis of Grid's Elements:

The analysis of grid elements is very similar to the grid segment analysis described in this document, so it is possible to use this document to analyze elements.

To display the results of the grid element analysis, select "Explorer" from the main menu, then click on the "Grid Element Statistics" icon on the right side of the map. In the Choose an analysis run dropdown select an available analysis run; any grid elements that have an analysis score for that run will have their icons on the map colored on a green-yellow-red scale; grid

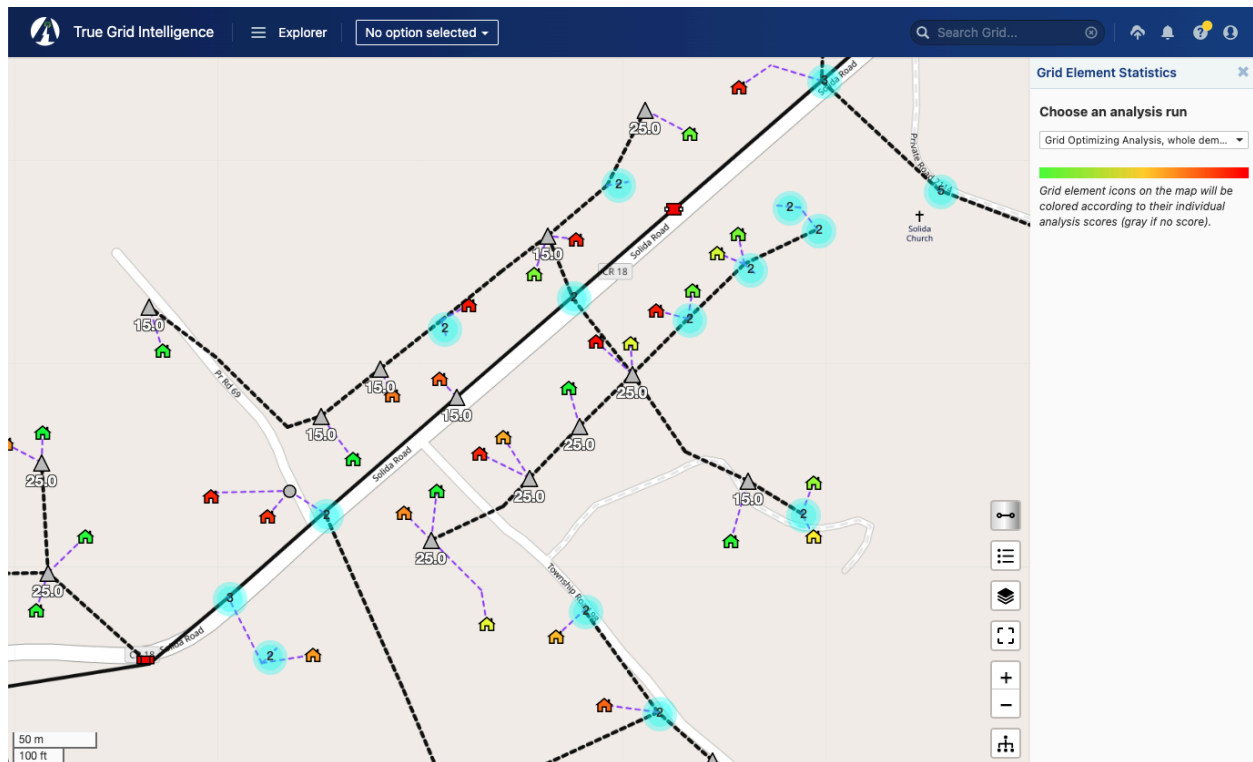
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elements without a score will have a gray icon. To revert to element coloring by phase, deselect the analysis run.



General Note

- This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different.
- Access to the different parts of TGI varies depending on the permissions of the user.
- Users on the Sandbox environments currently have read-only permissions, therefore, this use case may be accessible only for viewing, not creation or editing. If you are interested in the latter, please contact Awesense.
- Some of the data for this use case may be available only on Sandbox tiers 2 & 3, but not on tier 1. If you are interested in upgrading your tier access, please contact Awesense.
- For more information please contact tools@awesense.com.